📄 **Design Document (DD)**

🚑 **PinResQ – Geo-Verified Emergency Response System**

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1. 🧠 Introduction

This document defines the **system design, architecture, components, and UI structure** of the PinResQ web application.

It translates the **idea (from your PDF UI plan)** into a **technical implementation blueprint** using:

- React.js (Frontend)
- Spring Boot (Backend)
- MySQL (Database)

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2. 🎯 Design Goals

- Fast emergency reporting (≤ 5 seconds)
- Accurate GPS-based location capture
- Simple and intuitive UI (as per PDF screens)
- Scalable backend architecture
- Clear separation of concerns (Frontend ↔ Backend ↔ DB)

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3. 🧱 System Architecture

🔷 High-Level Architecture

[React Frontend]

↓ (Axios API Calls)

[Spring Boot Backend]

↓ (JPA/Hibernate)

[MySQL Database]

🔷 Component Architecture

Frontend Components

- Pages (Login, Home, Map, Rewards, Profile)
- API Service Layer
- Routing Layer

Backend Components

- Controller Layer (API endpoints)
- Service Layer (business logic)
- Repository Layer (DB interaction)
- Entity Layer (data models)

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4. 🎨 UI Design (Based on Your PDF)

Your PDF defines **5 main screens**, which map directly to React pages:

🏠 4.1 Home Page (Dashboard)

UI Elements:

- Emergency “Report” button
- Navigation to:
  - Map
  - Rewards
  - Profile

Functionality:

- Entry point after login

- Quick access to reporting

🗺️ 4.2 Map Page

UI Elements:

- Interactive map (Leaflet)
- Markers showing accident locations

Functionality:

- Fetch reports from backend
- Display markers
- Popup info on click

🎁 4.3 Rewards Page

UI Elements:

- Points display
- Reward history (future scope)

Functionality:

- Motivates users to report emergencies

👤 4.4 Profile Page

UI Elements:

- User details
- Toggle: **Volunteer Mode ON/OFF**

Functionality:

- Manage user info
- Enable volunteer participation

🔔 4.5 Notification Component

UI Elements:

- Alert popup / banner

Functionality:

- Notify nearby users (future real-time feature)

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5. 🔄 System Workflow

User → Login/Register

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Home Page

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Report Emergency

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Capture GPS Location

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Send to Backend API

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Store in MySQL

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Map Page Fetches Data

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Display Markers

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6. 🗄️ Database Design

Table: users

Field	Type
id	BIGINT
name	VARCHAR
email	VARCHAR
password	VARCHAR
phone	VARCHAR
volunteer_mode	BOOLEAN

Table: reports

Field	Type
id	BIGINT
latitude	DOUBLE
longitude	DOUBLE
timestamp	DATETIME
user_id	BIGINT

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7. ⚙️ Backend Design

🔹 Controller Layer

Handles API requests

Endpoints:

- POST /api/auth/register
- POST /api/auth/login
- POST /api/reports/create
- GET /api/reports/all

🔹 **Service Layer**

- Business logic
- Data processing

🔹 **Repository Layer**

- Extends JpaRepository
- Handles DB operations

🔹 **Entity Layer**

- Maps Java objects to database tables

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8. 🌐 Frontend Design

🔹 **Routing**

Route	Page
/	Login
/dashboard	Home
/map	Map
/rewards	Rewards
/profile	Profile

🔹 **State Management**

- React useState
- API calls via Axios

🔹 **API Integration**

Base URL:

http://localhost:8080/api

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9. 🗺️ Map Design (Leaflet Integration)

- Use OpenStreetMap tiles
- Center map on user location
- Add markers dynamically
- Popup shows:
  - "Emergency Reported"
  - Timestamp (optional)

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10. 🔐 Authentication Design

- Basic login/register (initial version)
- Store user session in localStorage
- Future:
  - JWT authentication

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11. 🔔 Notification Design (Future)

- Trigger when new report is created
- Show alert to nearby users
- Can be implemented using:
  - WebSockets
  - Firebase (optional)

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12. ⚠️ Error Handling

Frontend:

- Show alerts for API failures
- Handle empty data cases

Backend:

- Global exception handler
- Return structured error JSON

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13. 🧪 Testing Strategy

- API testing (Postman / browser)
- UI testing (React pages)
- Integration testing

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14. 🚀 Deployment Design (Optional)

- Frontend → Netlify / Vercel
- Backend → AWS / Render
- Database → MySQL server

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15. 🔮 Future Enhancements

- Image/video upload
- AI-based verification
- Real-time alerts
- Volunteer radius detection
- Mobile app version

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16. 🏁 Conclusion

This design document translates the **UI wireframe (your PDF)** into a **fully implementable system architecture**, ensuring:

- Clear mapping from design → development
- Scalable structure
- Easy implementation for beginners

✅ **You can use this for:**

- 📄 College Design Document submission
- 👨‍💻 Development reference
- 📊 PPT explanation
- 🧠 New chat for code generation